

Review 2 - Solutions

$$\textcircled{1} X \sim N(\theta, 4)$$

$$\text{prior } \theta \sim N(2, 2)$$

$$X = 2$$

posterior? MAP?

Bayes rule (CTS) case:

$$f(\theta|x) \propto \frac{f(x|\theta) \cdot f(\theta)}{\int f(x|\theta') \cdot f(\theta') d\theta'}$$

θ is mean of normal distribution $\Rightarrow$ its domain is $[-\infty, \infty]$

$$f(\theta|x) = \frac{\frac{1}{\sqrt{2\pi \cdot 4}} \exp\left(-\frac{(\theta-2)^2}{4}\right) \cdot \frac{1}{\sqrt{2\pi \cdot 2}} \exp\left(-\frac{(x-\theta)^2}{2}\right)}{\int \dots d\theta'}$$

$$= \text{const} \cdot \exp\left(-\frac{(\theta-2)^2}{4}\right) \cdot \exp\left(-\frac{(2-\theta)^2}{2}\right) =$$

$$= \text{const} \cdot \exp\left(-\frac{(\theta-2)^2(2+1)}{8}\right) = \text{const} \cdot \exp\left(-\frac{(\theta-2)^2}{2 \cdot 4/3}\right)$$

Since $f(\theta|x)$ is a density function, it must be normal (which is also $\text{const} \cdot \exp\left(-\frac{(\theta-\mu)^2}{2\sigma^2}\right)$)

$$\text{with } \mu = 2, \sigma^2 = 4/3$$

$$\text{So, posterior distribution is } \frac{1}{\sqrt{2\pi \cdot 4/3}} \cdot \exp\left(-\frac{(\theta-2)^2}{8/3}\right)$$

$$\boxed{\theta \sim N\left(2, \frac{4}{3}\right)}$$

$\oplus$ same mean as prior, but smaller var - we become more confident!

! In Bayesian problems, make sure twice before computing integral in the denominator that you actually need to compute it! It might take lots of time and be unnecessary!

MAP estimate $\hat{\theta} = \underset{\theta}{\operatorname{argmax}} f(\theta|x) =$

$$= \underset{\theta}{\operatorname{argmax}} \left(c \cdot \exp\left(-\frac{(\theta-2)^2}{8/3}\right) \right) = \textcircled{2}$$

at $\theta=2$ $\exp\left(-\frac{(\theta-2)^2}{8/3}\right) = 1$,
otherwise it's < 1

$$\boxed{\hat{\theta}=2}$$

gaussian density achieves
its max when $\theta = \mu$

② Linear regression model is $\alpha + (x_i - \bar{x})\beta + \epsilon_i$ $\epsilon_i = N(0, \sigma^2)$

You can either re-derive maximum likelihood estimates $\hat{\alpha}, \hat{\beta}, \hat{\sigma}^2$ or use formulas from the book:

$$\hat{\alpha} = \bar{y}$$

$$\hat{\beta} = \frac{\sum_{i=1}^n y_i (x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{\alpha} - \hat{\beta}(x_i - \bar{x}))^2$$

$$x_i = 32 \quad 23 \quad 23 \quad 30$$

$$y_i = 28 \quad 25 \quad 24 \quad 27$$

$$n=4$$

$$\bar{x} = \frac{1}{4} (32 + 23 + 23 + 30) = \frac{46 + 62}{4} = \frac{108}{4} = \textcircled{27}$$

$$\bar{y} = \frac{1}{4} (28 + 25 + 24 + 27) = \frac{80 + 13 + 11}{4} = \frac{104}{4} = \textcircled{26}$$

$$\hat{\alpha} = 26$$

$$x_i - \bar{x} = 5 \quad -4 \quad -4 \quad 3$$

$$\hat{\beta} =$$

$$\frac{28 \cdot 5 + 25(-4) + 24(-4) + 27 \cdot 3}{25 \cdot 4}$$

$$= \frac{140 - 100 - 96 + 81}{100} = \frac{25}{100} = \textcircled{\frac{1}{4}}$$

~~$$\frac{28 \cdot 5 + 25(-4) + 24(-4) + 27 \cdot 3}{25 \cdot 4}$$~~

$y = 26 + (x - 27) \cdot \frac{1}{4}$ - regression line

Thus, estimate for $X = 22$ is

$$y = 26 - 5 \cdot \frac{1}{2p} = 26 - \frac{5}{4} = 24.75$$

2²: if $x_i = 23$ $\hat{\alpha} + \hat{\beta}(x_i - \bar{x}) = 26 + (23 - 27) \cdot \frac{1}{4} = 26 - 1 = 25$

$$\begin{aligned} \text{if } x_i &= 30 \quad \hat{x} + \hat{\beta}(x_i - \bar{x}) = 26 + (30 - 27) \frac{1}{4} = \\ &= 26 + \frac{3}{4} = 27.25 \end{aligned}$$

$$x_i = 30 \quad 26 + \left(\frac{30 - 27}{4} \right) = 26 + \frac{3}{4} = 26.75$$

$$s^2 = \frac{1}{4} [(28 - 27.25)^2 + (25 - 25)^2 + (24 - 25)^2 + (27 - 26.75)^2] =$$

$$= \frac{1}{4} \cdot \left(1 \left(\frac{3}{4} \right)^2 + 0 + 1 + 1 \left(\frac{1}{4} \right)^2 \right) = \frac{9+16+1}{16 \cdot 4} = \frac{26}{64} = \frac{13}{32}$$

③ We would like to test the hypothesis that coin is fair
 $k=2$ (2 options, head or tail)

No: $P_1 = \frac{1}{2}, P_2 = \frac{1}{2}$ $n = 200$
 $P(\text{head})$ $P(\text{tail})$

$$s_{2-1}^2 = \sum_{i=1}^2 \frac{(y_i - n \cdot \frac{1}{2})^2}{n \cdot \frac{1}{2}} = \frac{(85-100)^2}{100} + \frac{(115-100)^2}{100} =$$

$$= \frac{2 \cdot 15^2}{100} = \frac{225.2}{100} = \textcircled{4.5}$$

Critical region $\{Z \geq q_{k-1}\}$

$$\chi_{0.05}^2(2-1) = \chi_{0.05}^2(1) \approx 3.8$$

Since $\{4.5 > 3.8\}$, we reject H_0 (coin is not fair)

p-value: ~~$\chi^2(1)$~~ $\textcircled{3.8}$

$$P(Z \geq q_{k-1})$$

where $Z \sim \chi^2(1)$

$$P(Z \geq q_{k-1}) = P(Z \geq 4.5) \approx 0.035$$

(between 0.05 and 0.025 in the table)

Since $p\text{-value} < 5\%$,
we reject H_0

$$\textcircled{4} f(x, \theta) = \frac{1}{\theta} x^{(1-\theta)/\theta}$$

Likelihood function is

$$L(\theta) = \prod_{i=1}^n f(x_i, \theta) = \left(\frac{1}{\theta}\right)^n \prod_{i=1}^n x_i^{(1-\theta)/\theta} = \frac{(\prod x_i)^{(1-\theta)/\theta}}{\theta^n}$$

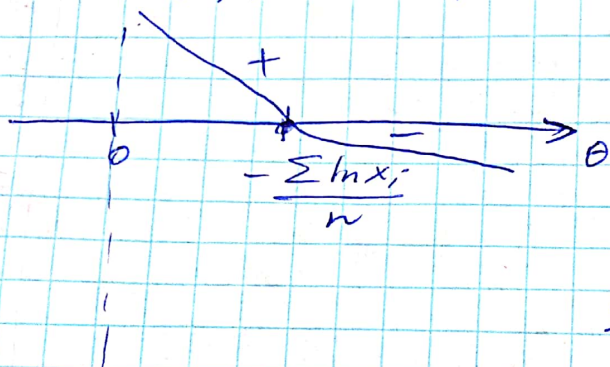
$$\ln L(\theta) = -n \ln \theta + \frac{1-\theta}{\theta} \left(\sum_{i=1}^n \ln x_i \right) \rightarrow \max ?$$

$$(\ln L(\theta))' = -\frac{n}{\theta} - \frac{1}{\theta^2} \left(\sum_{i=1}^n \ln x_i \right) = 0$$

$$\frac{(\theta n + \sum \ln x_i)}{\theta^2} = 0 \quad \theta = -\frac{\sum \ln x_i}{n} \quad \text{or } \theta = 0$$

Which one gives max?

$$x_i \in (0,1) \Rightarrow \ln x_i < 0 \Rightarrow \sum \ln x_i < 0$$



$$f(\theta) = \ln(L(\theta))' = \frac{-n\theta - \sum \ln x_i}{\theta^2}$$

$f(\theta) > 0$ for small θ

$f(\theta) < 0$ for large θ

$$\text{So, } \boxed{\hat{\theta} = -\frac{\sum \ln x_i}{n}}$$

$$b) \quad \mathbb{E} \hat{\theta} = -\frac{\sum_{i=1}^n \mathbb{E}(\ln x_i)}{n} =$$

$$= -\frac{\sum_{i=1}^n \frac{1}{\theta} \mathbb{E}(\ln(x_i))}{n} =$$

$$= -\frac{\sum_{i=1}^n \theta}{n} = \frac{n\theta}{n} = \theta$$

$\Rightarrow \hat{\theta}$ is an unbiased estimator

$$\mathbb{E} \ln X = \int_0^1 \underbrace{\frac{1}{\theta} t^{(1-\theta)/\theta}}_{v'} \underbrace{\ln t}_{u} dt =$$

$$= \left[\frac{1}{\theta} t^{1/\theta} \ln t \right]_0^1 - \int_0^1 t^{1/\theta} \cdot \frac{1}{t} dt =$$

$$= 0 - \int_0^1 t^{1/\theta - 1} dt = \left[\theta t^{1/\theta} \right]_0^1 = \theta$$

$$\left(\begin{array}{ll} v = t^{1/\theta} & v' = t^{1/\theta - 1} \cdot \frac{1}{\theta} \\ u = \ln t & u' = \frac{1}{t} \end{array} \right)$$